

Scrambled when wrong, left alone when right: a prediction-free account of learning in DishBrain

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This article responds to Kagan et al., “In vitro neurons learn and exhibit sentience when embodied in a simulated game-world,” *Neuron* **110**, 3952–3969 (2022).

Summary

Neurons in a dish learn to play Pong—but not, we argue, because they minimize surprise. The feedback after a miss is not only unpredictable; it is also twice as strong. We prove that this alone can teach: a network kicked at random when it fails, and left alone when it succeeds, drifts to where it fails least. No prediction, no reward. A single experiment can tell which account is true.

Main text

In DishBrain,¹ after a hit the culture receives a “predictable” stimulus: 100 Hz for 100 ms on all eight sensory electrodes at the sensory amplitude, 75 mV. After a miss it receives an “unpredictable” one: 150 mV at 5 Hz, at random sites and times, for 4 s, then 4 s without stimulation. Improvement under this feedback was interpreted as networks acting to make their input predictable, as the free-energy principle proposes.^{1,2} But the authors chose the higher voltage so that it would force action potentials “regardless of the state the cell was in, thereby being even more disruptive to the culture.”¹ The post-miss feedback is therefore both less predictable and more disruptive, and the published controls do not separate the two: *No feedback* removes both differences, and *Silent* still withdraws the sensory input for eight seconds after a miss only. Earlier discussion of the work concerned the terms “intelligence” and “sentience”;^{3,4} our question is which property of the feedback teaches.

The scrambler

Let ψ be the network’s setting and $P(\psi)$ its probability of missing. Suppose a miss moves ψ by a symmetric kernel $K(\psi, \psi') = K(\psi', \psi)$ and a hit leaves it in place. Nothing tells the network which way to change.

Proposition 1. *If $P > 0$ and K is irreducible, the stationary distribution is $\rho(\psi) \propto 1/P(\psi)$. The long-run miss rate is the harmonic mean of P , below its arithmetic mean (the miss rate of a random setting) unless P is constant.*

A setting that misses ten times less often is kept ten times longer: good settings are not sought, only left alone. In general both outcomes perturb, with variance σ_m^2 after a miss and σ_h^2 after a hit; for small isotropic steps the setting diffuses with $D(\psi) = \frac{1}{2}[\sigma_m^2 P + \sigma_h^2(1 - P)]$, and the stationary density is $\rho \propto 1/D$.⁵

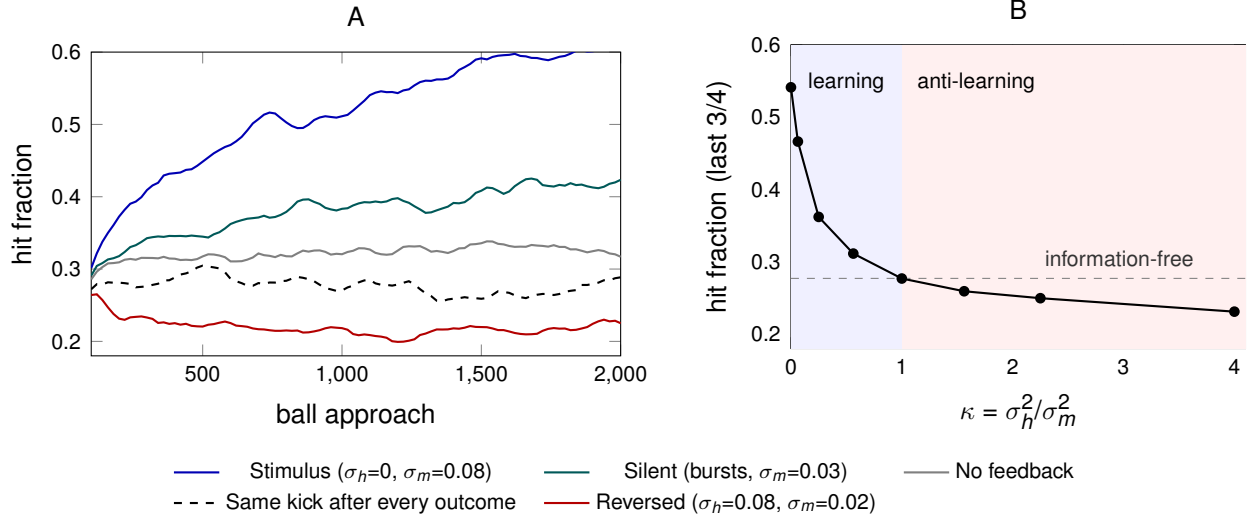


Figure 1. Learning by outcome-contingent perturbation in a closed-loop Pong model. (A) Hit fraction along a session (sliding window of 100 approaches; mean of 400 simulated cultures). (B) Hit fraction over the last three quarters of a session against the ratio κ of perturbation variances after a hit and after a miss ($\sigma_m = 0.08$); dashed: the information-free level. Nothing in the model computes a prediction, a reward or the sign of an error.

Proposition 2. With $\kappa = \sigma_h^2 / \sigma_m^2$, $\rho \propto 1/[\kappa + (1 - \kappa)P]$. The stationary miss rate is below the uniform average for $\kappa < 1$ (learning), equal to it for $\kappa = 1$, and above it for $\kappa > 1$ (anti-learning).

Proofs are in the STAR Methods. Predictability does not enter; what matters is how far each outcome moves the network. The principle is old—Ashby’s ultrastability,⁶ orthokinesis,⁵ the stimulus regulation principle in cultures^{7,8} and learning by stimulation avoidance^{9,10}—but none of these works analyzes DishBrain. We add that DishBrain’s own feedback contains this contingency, its quantitative consequences including the sign reversal at $\kappa = 1$, and experiments that decide.

A closed-loop model

On a two-parameter paddle model (31×21 grid of settings) Proposition 1 predicts a stationary miss rate of 0.815; simulation gives 0.810, against 0.845 for a random setting. In a DishBrain-like loop the ball is place- and rate-coded on 8 channels as in the experiment, two motor populations read it through 16 plastic weights, and the only teacher is a random kick to all weights after each approach. Silent is modeled as a weaker scrambler ($\sigma_m = 0.03$ against 0.08), because unstimulated cultures burst¹¹ and bursts make weights drift;¹² this is a testable assumption.

Over 400 simulated cultures the within-session gain in hit fraction is $+0.161 \pm 0.014$ (mean $\pm$ SEM) for Stimulus, $+0.068 \pm 0.012$ for Silent and $+0.017 \pm 0.007$ for No feedback (Figure 1A): the reported ordering, with Silent above No feedback but weaker.¹ The same kick after every outcome gives -0.010 ± 0.007 , and reversed disruption -0.019 ± 0.006 . Performance falls monotonically with κ and crosses the information-free level at $\kappa = 1$ (Figure 1B). The model is not fitted to data; the claim is only that the contingency suffices.

Protocol	After a hit	After a miss	Predictability	Scrambler
A. Published	75 mV, 100 Hz, 100 ms	150 mV random, 4 s; 4 s without input	learns	learns
B. Predictable penalty	as A	150 mV, same dose as A, in a fixed sequence	weak or none	learns as A
C. Quenched penalty	as A	8 s of random rapid low-amplitude stimulation that suppresses bursts ¹¹	learns	weak or none
D. Reversed	150 mV fixed, 1 s	75 mV random, 1 s	learns	below E
E. Information-free	same random stimulus	same random stimulus	none	none

Table 1. Protocols that separate the accounts. B and C change one property of the post-miss feedback and hold the other; D tests the sign rule of Proposition 2; E is the reference level for D.

Experiments that decide

The accounts agree on every condition tested so far and disagree on those in Table 1. Two analyses of existing recordings also decide. *Per-event displacement*: a miss should move the network's state (e.g., the evoked response to a fixed probe) further than a hit, and across cultures the gain should grow with this difference. *Bursts after a miss*: in Silent, the number of spontaneous bursts after misses should predict each culture's gain.

If the scrambler account holds, DishBrain shows a generic property of plastic networks under outcome-contingent perturbation rather than predictive processing, though a surprise-minimizing system may implement exactly this. It also explains why gains did not carry over between sessions:¹ a setting held only by the absence of perturbation is diffused by bursting.¹² A limitation is that stimulation is not isotropic, and a 100 Hz burst may itself potentiate specific pathways;¹³ the displacement analysis measures this directly.

Acknowledgments

[Optional.]

Author contributions

K.K. conceived the study, performed the analysis and simulations, and wrote the paper.

Declaration of interests

The author declares no competing interests.

STAR Methods

Resource availability

Lead contact. Requests for further information should be directed to the lead contact, Konstantin Kladko ([email]).

Materials availability. This study did not generate new materials.

Data and code availability. This paper analyzes no new experimental data. The simulation code (scrambler.py, standard-library Python) and the data behind Figure 1 are publicly available at <https://github.com/kladkogex/20-watt-computer> (folder papers/dishbrain-scrambler). Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

Method details

Proof of Proposition 1. Between approaches the setting moves from ψ to ψ' with probability $P(\psi)K(\psi, \psi')$. The measure $\rho \propto 1/P$ satisfies detailed balance, $\rho(\psi)P(\psi)K(\psi, \psi') = \rho(\psi')P(\psi')K(\psi', \psi)$, because K is symmetric, and is unique since K is irreducible and $P > 0$. The long-run miss rate $\sum \rho P = |\Omega| / \sum 1/P$ is the harmonic mean of P , at most the arithmetic mean.

Proof of Proposition 2. With the step size set at the point of departure, the density obeys $\partial_t \rho = \nabla^2(D\rho)$; with a reflecting boundary the stationary flux vanishes, so $\rho \propto 1/D$. Writing $w = 1/[\kappa + (1 - \kappa)P]$ and $\mathbb{E}_U$ for the uniform average, the miss rate is $\mathbb{E}_U[Pw]/\mathbb{E}_U[w]$. For $\kappa < 1$, w decreases with P and Chebyshev's association inequality gives $\mathbb{E}_U[Pw] \leq \mathbb{E}_U[P]\mathbb{E}_U[w]$; for $\kappa > 1$ the inequality reverses.

Closed-loop model. Court 1×1 ; paddle half-length 0.1 on the right wall; steps of 20 ms. Each serve starts at the far wall with horizontal speed 1 and vertical speed uniform in $[-0.8, 0.8]$. The ball's height relative to the paddle selects one of 8 channels, and its position x sets the rate $s = (4 + 36x)/40$. Motor rates are $[W_{\uparrow, \downarrow}[k]s + \xi]_+$ with Gaussian noise (SD 0.35 per step), and the paddle moves by $1.5(r_{\uparrow} - r_{\downarrow})$ court heights per second. The 16 weights start uniform in $[-1, 1]$ and reflect at ± 1 . After every approach each weight receives a Gaussian kick of SD 0.01, then one of SD σ_h after a hit or σ_m after a miss. Conditions (σ_h, σ_m) : Stimulus (0, 0.08), Silent (0, 0.03), No feedback (0, 0), same kick (0.08, 0.08), reversed (0.08, 0.02).

Quantification and statistical analysis

Each of 400 simulated cultures plays one session of 2000 ball approaches. The within-session gain is the hit fraction over approaches 501–2000 minus that over approaches 1–500; values are mean $\pm$ SEM over cultures. The grid check of Proposition 1 runs the chain for 3×10^6 approaches.

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